

YIRAN HU

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SUMMARY

- UC Berkeley **Math CS** student (**3.97** GPA) experienced in **AI**, **math research**, **RAG**
- Co-Founder & CTO @ **Ambees**, developed personalized **Agentic AI** and **RAG** system tailored to each user
- Research in **LLM benchmark**, **mathematics**, and **AI policy**, ship products in **SwiftUI**, **Next.js**, including award-winning **iOS app**

EDUCATION

University of California, Berkeley Expected May 2029
B.A. in Mathematics and Computer Science, GPA: 3.97 Berkeley, CA
Awards: Cum Laude Society, John Suydam Mathematics Prize, USAMO 2026 Grader
Coursework: AI, Data Structures, Lin Alg & Diff Eqs, Discrete Math & Prob, Real Analysis, Data Science, Honors Multivariable Calculus

ACADEMIC RESEARCH

Data Discovery Program | LLM Tutors for All, UC Berkeley Data Science Undergraduate Studies Sep 2026 – Present
Project Lead: Edwin Vargas Navarro Berkeley, CA

- Benchmarking course-deployed LLM tutors (**Data 8**, **Data 100**, **E127**) on mistake identification, answer withholding, and actionability
- Building **Python** benchmark comparing providers, **RAG** vs. raw **PDF** context, **PDF-to-Markdown**, and **token cost**

Data Discovery Program | UC Berkeley Data Science Undergraduate Studies Jan 2026 – May 2026
Advisor: Eric Van Dusen Berkeley, CA

- Benchmarking small laptop-runable models (Qwen 2.1, TinyLlama) using **Python**, **HuggingFace Transformers**, **llama.cpp**
- Build **Jupyter** pipelines for accuracy, bias, and compute tradeoffs; design **RAG** curriculum for future Berkeley course.

Research Assistant | UC Berkeley Department of Mathematics Jan 2026 – May 2026
Advisor: Prof. Alan Hammond Berkeley, CA

- Investigating **Nash equilibria** in **Brownian Boost** tug-of-war games with **resource allocation** dynamics
- Built and deployed an interactive finite **TLP game** and analyzed **human** play traces with **ABMN equations**.

Undergraduate Research Apprentice | UC Berkeley School of Education Sep 2025 – Present
Mentor: Prof. Jose Eos Trinidad | Donor-funded \$3,500 summer research stipend Berkeley, CA

- Co-authoring two papers in preparation for *Sociology Compass* and *Organization Theory*
- Drafted section synthesizing **30+ empirical studies** on **organizational structure**
- Building cross-sector case base for four **human-AI work arrangements** (tool, assistant, co-worker, supervisor)
- Coauthored **27-page policy paper** analyzing how 12 large U.S. school districts govern **generative AI**

EXPERIENCE

Co-Founder & CTO Aug 2025 – Present
Ambees Berkeley, CA

- Built a full-stack early-career recruiting platform using **Next.js**, **React**, **Neon PostgreSQL**, and **Vercel**
- Developed an AI voice onboarding and multi-stage job matching system using **OpenAI Realtime**
- Engineered resumable application automation for **Greenhouse**, **Lever**, **Ashby**, and **Workday** using **Playwright** and **Browserbase**
- Completed **Summer Launch Incubator (SLI)**, refining product through customer discovery, venture mentorship, iterative validation

Senior CS 88 Mentor | CSM Sep 2025 – Present
UC Berkeley Berkeley, CA

- Lead weekly review sessions for CS 88 students covering **Python**, **recursion**, and **data abstraction** concepts
- Create video walkthroughs for past exam problems benefiting current and future cohorts

PERSONAL PROJECTS & HACKATHONS

bloop | *Swift, SwiftUI, HealthKit, Google Gemini API, WidgetKit* | **HoyaHacks 3rd Place + Best Domain (bloopy.tech)**

- Built a privacy-first iOS app using **HealthKit** to track headphone sound exposure and protect children's hearing health
- Integrated **Gemini-powered agentic AI** for listening-pattern analysis; added dose visualization, alerts, Live Activities, and **widgets**

MindJournal | *Next.js, React, TypeScript, D3.js, OpenAI API, PostgreSQL* | mindjournal.org

- Built **AI-powered reflective journaling platform** with Socratic dialogue and cognitive-graph **visualization** of recurring themes
- Designed insight summaries and weekly progress tracking with **OpenAI integration** for personalized growth feedback

TalkStorm | *JavaScript, Groq API, MindElixir, Real-time Processing*

- Built **real-time Groq agentic-AI** mind map generator for lectures with heap-based topic prioritization and sub-second latency

SKILLS

Technical Skills: Java, Python, JavaScript, SQL, REST APIs, RAG, PyTorch, NumPy, Databases, SwiftUI, L^AT_EX, Git, Jupyter, AI Agent